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Oefeningenreeks 4A nr. 14.3

$$\begin{aligned} & \int (x^2 + x + 2) \ln(x) dx \\ & \left\{ \begin{array}{l} u = \ln(x) \\ dv = (x^2 + x + 2) dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = \frac{1}{x} dx \\ v = \frac{x^3}{3} + \frac{x^2}{2} + 2x \end{array} \right. \\ & = \left(\frac{x^3}{3} + \frac{x^2}{2} + 2x \right) \ln(x) - \int \left(\frac{x^3}{3} + \frac{x^2}{2} + 2x \right) \frac{1}{x} dx \\ & = \left(\frac{x^3}{3} + \frac{x^2}{2} + 2x \right) \ln(x) - \int \left(\frac{x^2}{3} + \frac{x}{2} + 2 \right) dx \\ & = \left(\frac{x^3}{3} + \frac{x^2}{2} + 2x \right) \ln(x) - \left(\frac{x^3}{9} + \frac{x^2}{4} + 2x \right) + c \end{aligned}$$

References